

Introduction to Machine Learning

HOMEWORK 2

Linear-regression Models for Housing Price

US Housing Dataset

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[GitHub repository](#)

1. Methodology and Assumptions

The objective of this homework was to develop linear-regression models for housing price using a manually implemented gradient-descent algorithm. The dataset contains 545 observations and 13 columns. The target variable is price. The implementation used NumPy for the gradient-descent calculations and scikit-learn only for the 80/20 split and the allowed feature-scaling operations.

1.1 Assumptions

- An 80% training and 20% held-out evaluation split was used for every problem, with `random_state = 100` so that all comparisons used the same observations.
- The assignment uses the terms evaluation/test and validation for the held-out 20%. In this report, it is called the validation set because that naming is used in the implementation.
- Binary yes/no variables were encoded as 1/0. The `furnishingstatus` variable was not used because it was not included in the feature lists specified in the problems.
- Problem 1 used raw, unscaled inputs as instructed.
- Problem 2 scaled only the input variables. Price was not normalized or standardized.
- For Problem 3, “parameter penalty” was implemented as L2 regularization. The intercept was not penalized, and the validation loss remained ordinary MSE.

1.2 Gradient-descent model

A bias column of ones was added to the input matrix so that the intercept could be learned as part of the parameter vector. All parameters were initialized to zero, as required. For Problems 1 and 2, the mean-squared-error objective and its batch gradient were:

$$J(\theta) = (1/m) \sum (\hat{y}_i - y_i)^2$$

$$\nabla J(\theta) = (2/m) X^T(X\theta - y)$$

$$\theta \leftarrow \theta - \alpha \nabla J(\theta)$$

Training data were used to update the parameters. The held-out data were used only to calculate the validation MSE. The main experiments used 1,000 iterations; the additional raw-feature stability experiment in Problem 1 used 10,000 iterations because the learning rate had to be extremely small.

2. Problem 1(a): Five Unscaled Input Features

The first model used area, bedrooms, bathrooms, stories, and parking to predict price. No normalization or standardization was applied.

2.1 Required learning-rate exploration

Learning rates of 0.01, 0.05, and 0.10 were tested on the raw features. The loss increased explosively in every case, so meaningful fitted parameters were not obtained from the required range.

Learning rate	Final training MSE	Final validation MSE
0.01	1.7252×10^{129}	1.7635×10^{129}
0.05	1.6453×10^{143}	1.6818×10^{143}
0.10	1.7252×10^{149}	1.7635×10^{149}

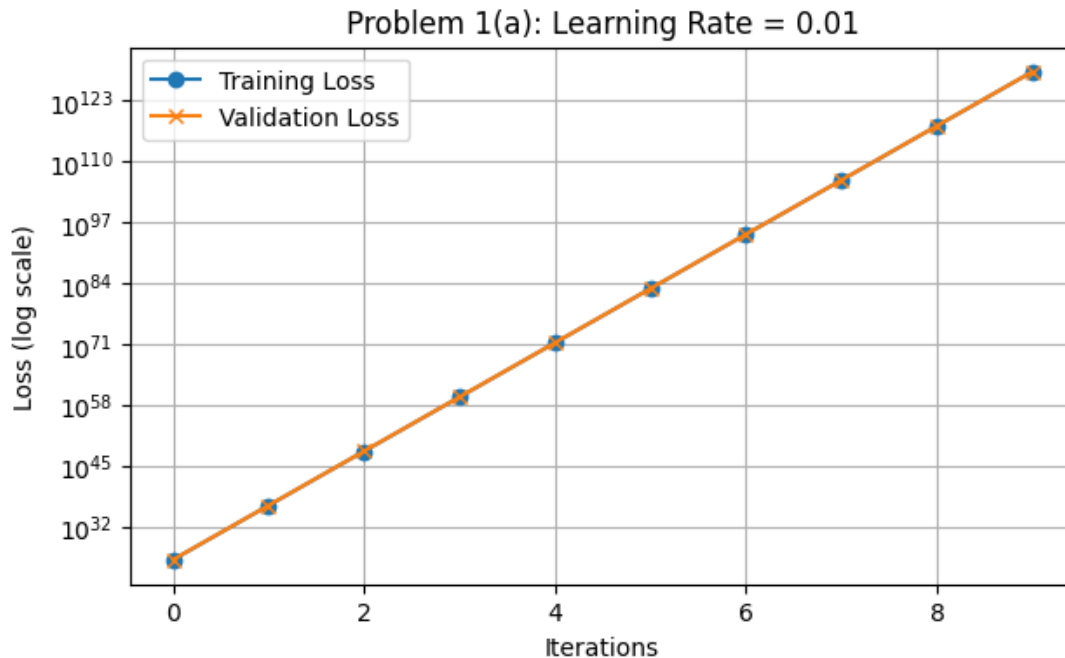


Figure 1. Problem 1(a), $\alpha = 0.01$: loss diverges on the unscaled features.

The area variable is measured in the thousands, whereas bedrooms, bathrooms, stories, and parking are small counts. Consequently, the raw gradients have very different magnitudes. With $\alpha = 0.01$ – 0.10 , the update step is too large and gradient descent overshoots instead of approaching a minimum. Increasing α makes the divergence more severe.

2.2 Additional stability experiment

Following the instructor's clarification, a much smaller learning rate of 10^{-9} was also tested for 10,000 iterations. This produced stable, decreasing losses, although convergence was slow.

Quantity	Result
Learning rate	1×10^{-9}
Iterations	10,000
Training MSE	3.38005×10^{12}
Validation MSE	3.21879×10^{12}

Parameter	Feature	Value
θ_0	Intercept	7.3604
θ_1	Area	861.8880
θ_2	Bedrooms	27.6745
θ_3	Bathrooms	16.1044
θ_4	Stories	24.7364
θ_5	Parking	5.7621

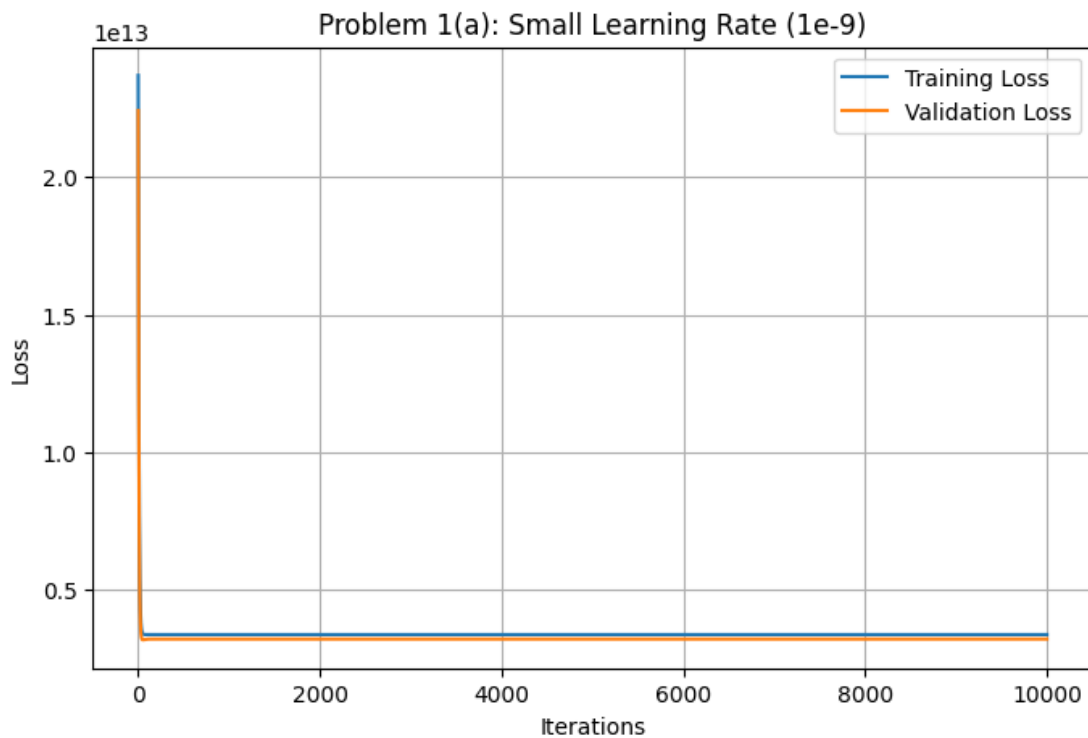


Figure 2. Problem 1(a): stable but slow convergence with $\alpha = 10^{-9}$.

These parameters are reported as the values obtained after 10,000 iterations, not as a claim of full convergence. The experiment demonstrates why feature scaling becomes important in Problem 2.

3. Problem 1(b): Eleven Unscaled Input Features

The second raw-input model added mainroad, guestroom, basement, hotwaterheating, airconditioning, and prefarea to the five features from Problem 1(a). The six yes/no variables were encoded as 1 and 0.

Learning rate	Final training MSE	Final validation MSE
0.01	1.7252×10^{129}	1.7635×10^{129}
0.05	1.6453×10^{143}	1.6818×10^{143}
0.10	1.7252×10^{149}	1.7635×10^{149}

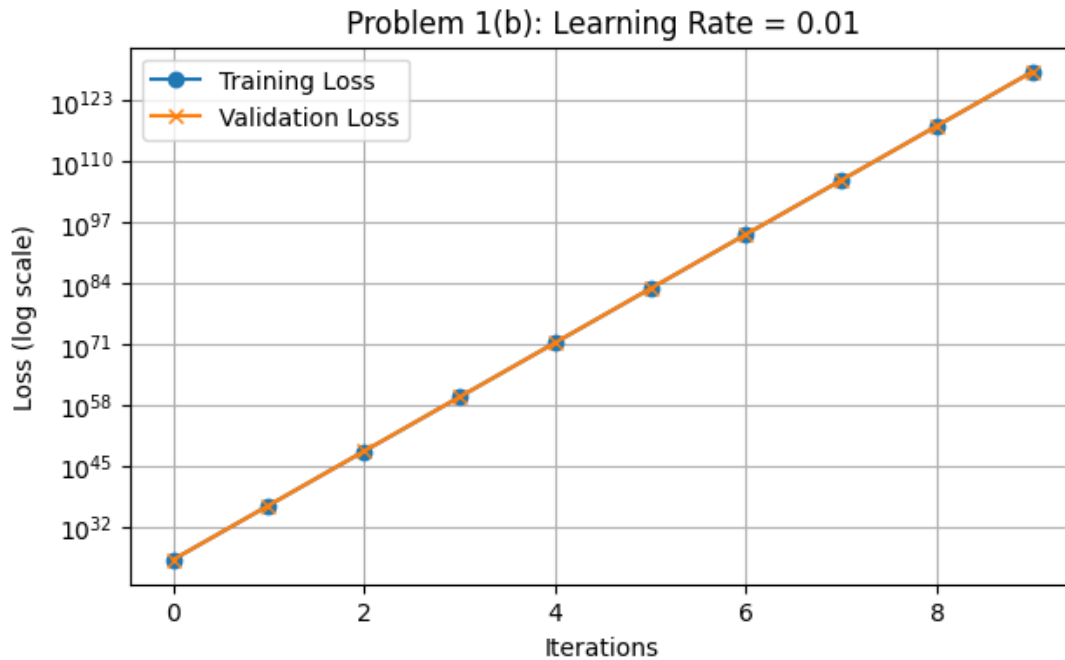


Figure 3. Problem 1(b), $\alpha = 0.01$: the unscaled 11-feature model also diverges.

The same divergence occurs because adding binary variables does not remove the large scale difference created by area. The required learning-rate range therefore does not produce meaningful final parameters.

3.1 Additional small-learning-rate run

Quantity	Problem 1(a)	Problem 1(b)
Training MSE	3.380050×10^{12}	3.380039×10^{12}
Validation MSE	3.218790×10^{12}	3.218776×10^{12}

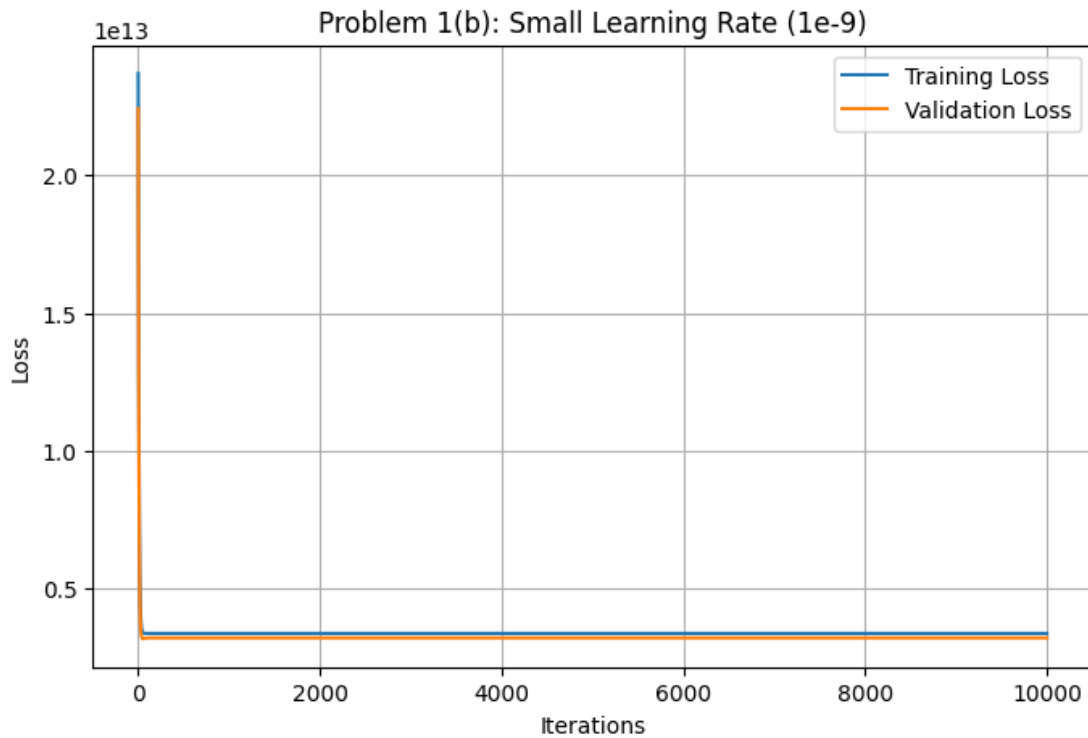


Figure 4. Problem 1(b): stable but slow convergence with $\alpha = 10^{-9}$.

At $\alpha = 10^{-9}$, Problem 1(b) was stable but only slightly lower in MSE than Problem 1(a). Because the raw variables are on very different scales, the small-scale coefficients learn very slowly. This result should therefore be viewed as a stability demonstration rather than a fully converged comparison of feature usefulness.

4. Problem 2(a): Scaling the Five-Feature Model

Problem 2(a) repeated Problem 1(a) using two preprocessing methods: Min-Max normalization and standardization. Each scaler was fitted on the training data only and then applied to the validation data. Price remained in its original units.

4.1 Min-Max normalization

The training features were mapped to the interval $[0,1]$. Learning rates from 0.01 through 0.10 were tested for 1,000 iterations.

α	Training MSE	Validation MSE
0.01	1.7732×10^{12}	1.5741×10^{12}
0.02	1.6256×10^{12}	1.5094×10^{12}
0.03	1.5677×10^{12}	1.5154×10^{12}
0.05	1.5317×10^{12}	1.5495×10^{12}
0.10	1.5224×10^{12}	1.5874×10^{12}

Among the tested normalization runs, $\alpha = 0.02$ gave the lowest validation MSE and was retained for the plotted normalization result.

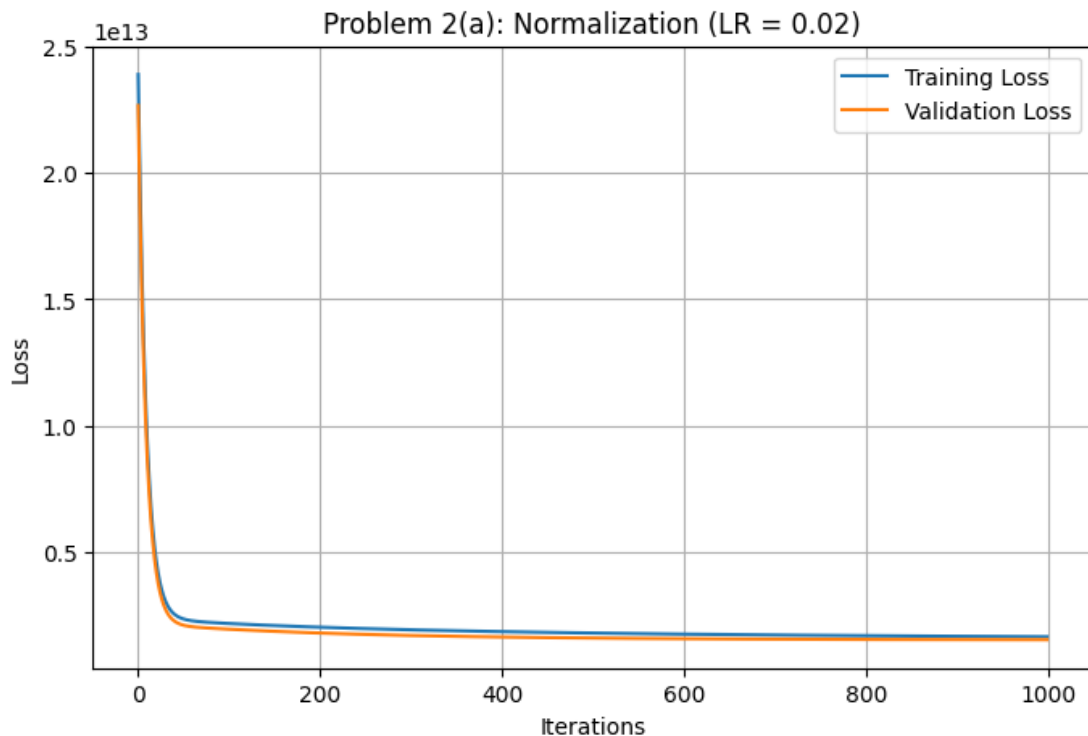


Figure 5. Problem 2(a): Min-Max normalization with $\alpha = 0.02$.

4.2 Standardization

Standardization transformed each training feature to approximately zero mean and unit standard deviation. The learning-rate trials converged rapidly; α values of 0.02 and above reached essentially the same training minimum within 1,000 iterations. A representative converged run with $\alpha = 0.05$ was used.

Method	Selected α	Training MSE	Validation MSE
Normalization	0.02	1.6256×10^{12}	1.5094×10^{12}
Standardization	0.05	1.5221×10^{12}	1.5965×10^{12}

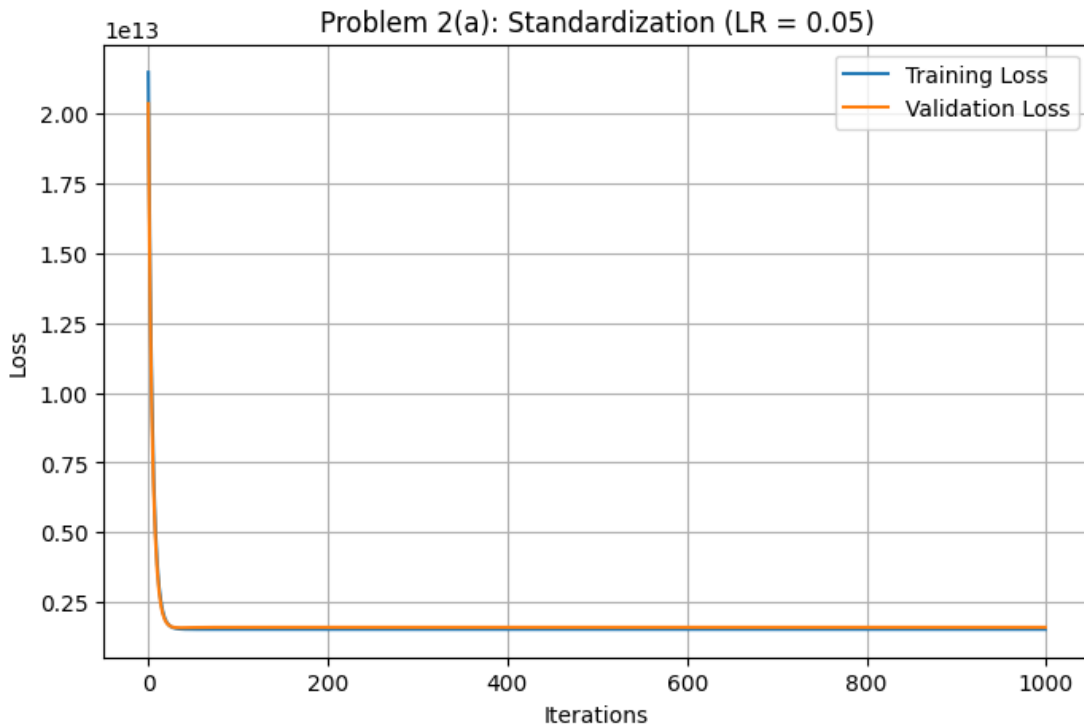


Figure 6. Problem 2(a): standardization with $\alpha = 0.05$.

Compared with Problem 1(a), both scaling methods prevented numerical divergence at the required learning-rate scale. For the selected 1,000-iteration experiments, standardization achieved the lower training MSE and reached its plateau much faster. Normalization at $\alpha = 0.02$ had the lower validation MSE in the reported selected runs. Because both transformations are invertible affine rescalings and this model is unregularized, they do not create different least-squares problems when optimization is carried to full convergence; the observed difference mainly reflects convergence speed and the selected finite-iteration runs.

5. Problem 2(b): Scaling the Eleven-Feature Model

Problem 2(b) repeated the scaling comparison using all 11 specified predictors.

5.1 Min-Max normalization

The validation MSE decreased through $\alpha = 0.03$ and then began to increase. Therefore $\alpha = 0.03$ was selected from the tested normalization learning rates.

α	Training MSE	Validation MSE
0.01	1.3183×10^{12}	1.1757×10^{12}
0.02	1.2183×10^{12}	1.1138×10^{12}
0.03	1.1806×10^{12}	1.1127×10^{12}
0.05	1.1561×10^{12}	1.1315×10^{12}
0.10	1.1488×10^{12}	1.1518×10^{12}

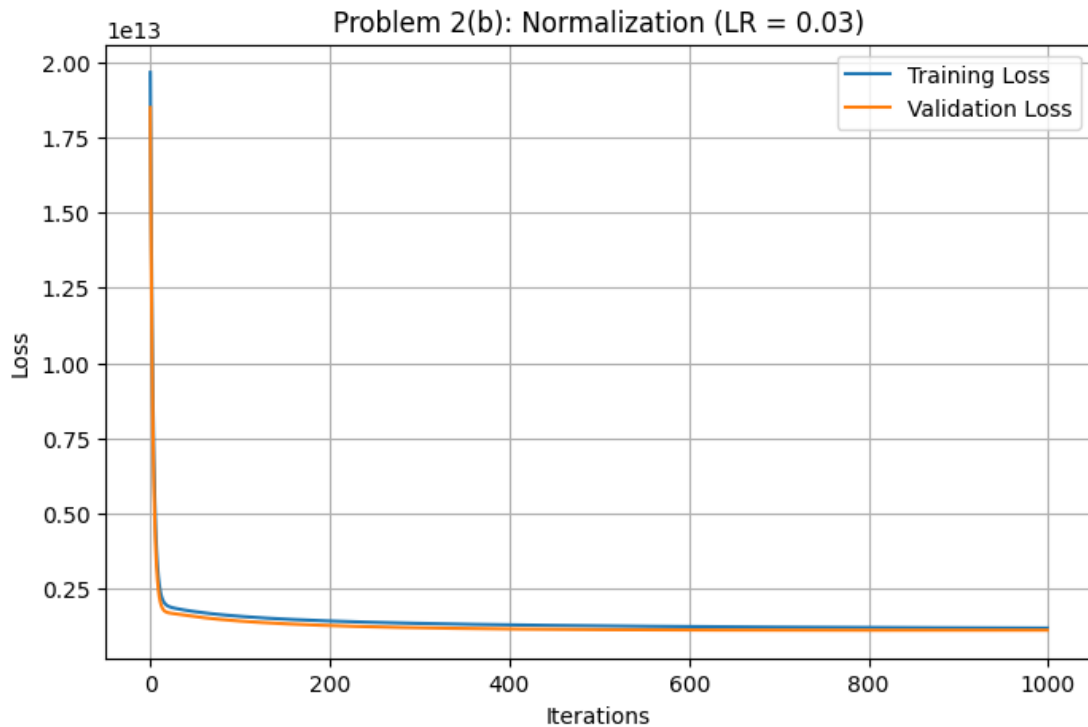


Figure 7. Problem 2(b): Min-Max normalization with $\alpha = 0.03$.

5.2 Standardization and comparison

The standardized model again converged rapidly. A representative $\alpha = 0.05$ run reached a training MSE of 1.14847×10^{12} and validation MSE of 1.15591×10^{12} .

Method	Selected α	Training MSE	Validation MSE
Normalization	0.03	1.1806×10^{12}	1.1127×10^{12}
Standardization	0.05	1.1485×10^{12}	1.1559×10^{12}

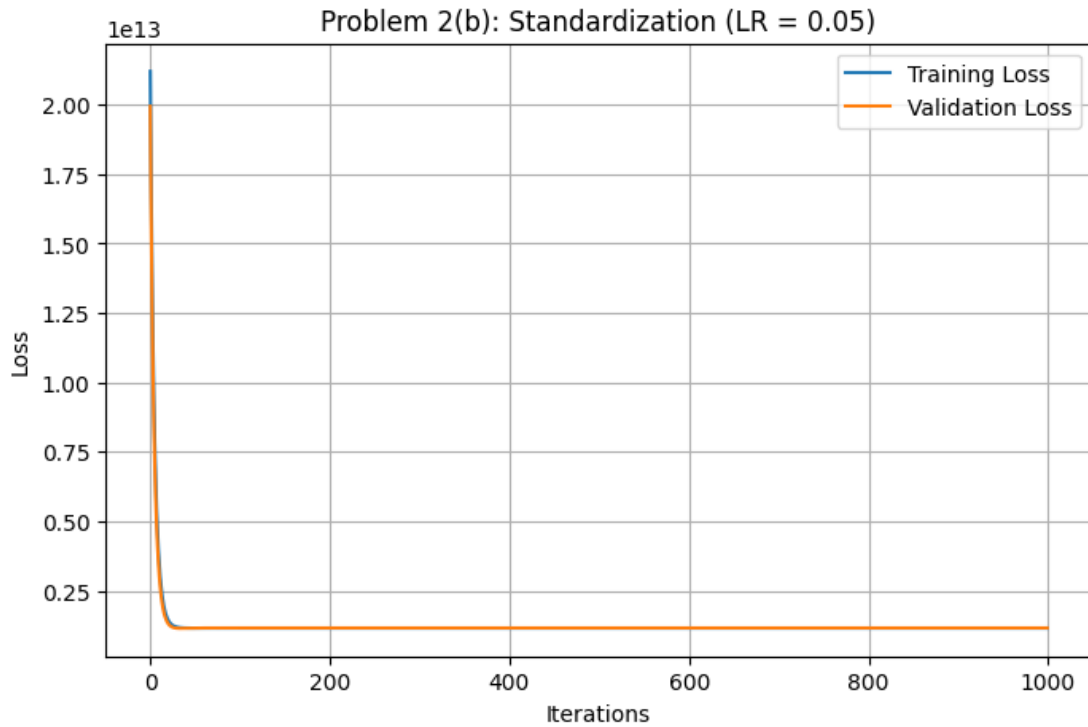


Figure 8. Problem 2(b): standardization with $\alpha = 0.05$.

For the fixed 1,000-iteration comparison requested by the assignment, standardization produced the lower training MSE and faster convergence, while the selected normalization run produced the lower validation MSE. Both methods eliminated the divergence seen in Problem 1(b). Adding the six additional housing indicators also produced lower held-out error in these experiments than the corresponding five-feature models.

6. Problem 3(a): Five Features with L2 Parameter Penalty

Problem 3 adds a parameter penalty to the training objective. L2 regularization was assumed. Standardization was used because it was the best scaling approach by the training-performance criterion in Problem 2(a). The regularized objective was:

$$J_{\text{reg}}(\theta) = (1/m) \sum (\hat{y}_i - y_i)^2 + (\lambda/m) \sum \theta_j^2, \quad j = 1, \dots, n$$

The intercept θ_0 was excluded from the penalty. Accordingly, the regularization gradient for $\theta_1 \dots \theta_n$ was $(2\lambda/m)\theta$. The validation loss was not regularized, following the assignment instruction.

6.1 Lambda search

λ	Regularized training loss	Validation MSE
0	1.5221×10^{12}	1.5965×10^{12}
10	1.5502×10^{12}	1.5892×10^{12}
50	1.6537×10^{12}	1.5732×10^{12}
75	1.7121×10^{12}	1.5713×10^{12}
100	1.7664×10^{12}	1.5739×10^{12}
200	1.9514×10^{12}	1.6131×10^{12}
500	2.3185×10^{12}	1.8202×10^{12}

The broad search located the useful region, and the refined search over $\lambda = 50, 75, 100, 125$, and 150 identified $\lambda = 75$ as the lowest validation MSE among the tested values.

6.2 Final model and comparison with Problem 2(a)

Parameter	Feature	2(a), $\lambda=0$	3(a), $\lambda=75$
θ_0	Intercept	4,795,729	4,795,729
θ_1	Area	729,932	636,030
θ_2	Bedrooms	79,335	124,392
θ_3	Bathrooms	641,265	564,722
θ_4	Stories	463,856	406,823
θ_5	Parking	287,331	277,715

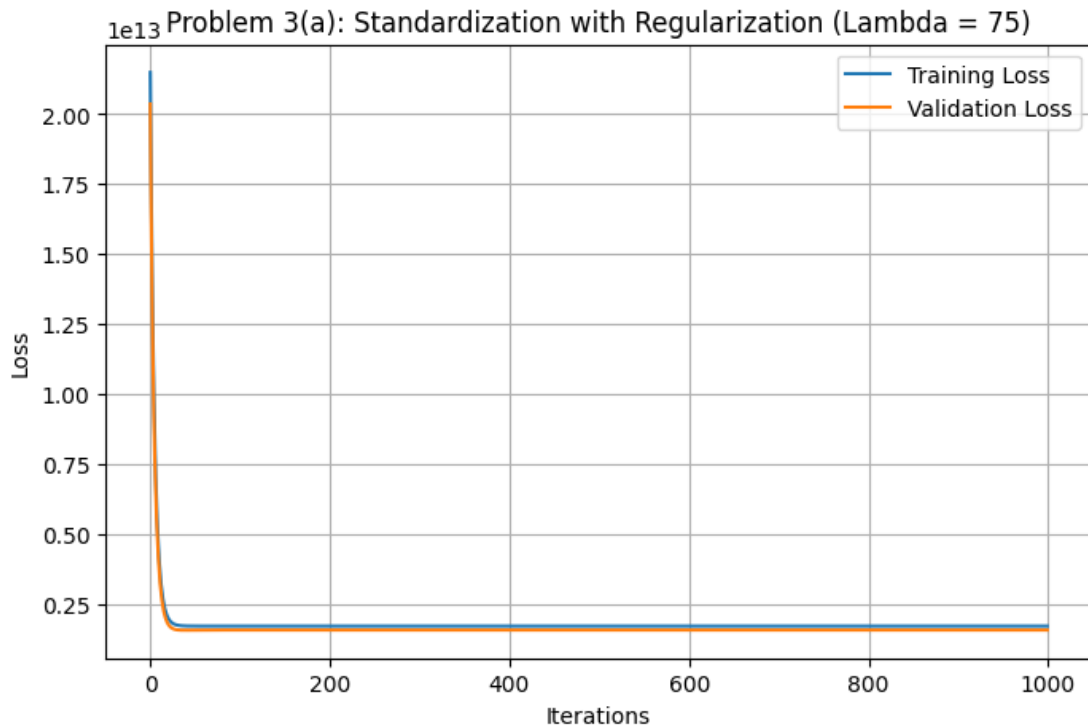


Figure 9. Problem 3(a): standardized five-feature model with L2 regularization, $\lambda = 75$.

Validation MSE decreased from 1.59653×10^{12} without regularization to 1.57129×10^{12} with $\lambda = 75$, a reduction of approximately 1.58%. Several coefficients became smaller in magnitude, illustrating the shrinkage effect of L2 regularization. Individual coefficients do not have to shrink monotonically in a correlated multivariable model, as seen for bedrooms. Larger λ values eventually increased validation error, indicating underfitting from excessive regularization.

7. Problem 3(b): Eleven Features with L2 Parameter Penalty

Problem 3(b) applied the same L2 formulation to the standardized 11-feature model from Problem 2(b), using $\alpha = 0.05$ and 1,000 iterations.

7.1 Lambda search

λ	Regularized training loss	Validation MSE
0	1.1485×10^{12}	1.1559×10^{12}
10	1.1748×10^{12}	1.1515×10^{12}
50	1.2731×10^{12}	1.1419×10^{12}
65	1.3074×10^{12}	1.14107×10^{12}
70	1.3186×10^{12}	1.141069×10^{12}
75	1.3297×10^{12}	1.14120×10^{12}
100	1.3830×10^{12}	1.1436×10^{12}
200	1.5702×10^{12}	1.1754×10^{12}
500	1.9672×10^{12}	1.3605×10^{12}

A broad search first showed that the minimum was near $\lambda = 50$ – 100 . A refined search from 55 to 90 in increments of 5 gave the lowest observed validation MSE at $\lambda = 70$. The result at $\lambda = 65$ was extremely close, so $\lambda = 70$ is reported as the best value among the tested candidates rather than as a universal optimum.

7.2 Final parameters

Parameter	Feature	2(b), $\lambda=0$	3(b), $\lambda=70$
00	Intercept	4,795,729	4,795,729
01	Area	532,836	478,781
02	Bedrooms	72,722	110,032
03	Bathrooms	578,492	514,154
04	Stories	367,613	331,379
05	Main road	210,296	205,424
06	Guest room	146,049	150,984
07	Basement	145,416	128,938
08	Hot-water heating	204,959	175,921
09	Air conditioning	399,907	368,760
010	Parking	222,498	220,370
011	Preferred area	255,724	237,744

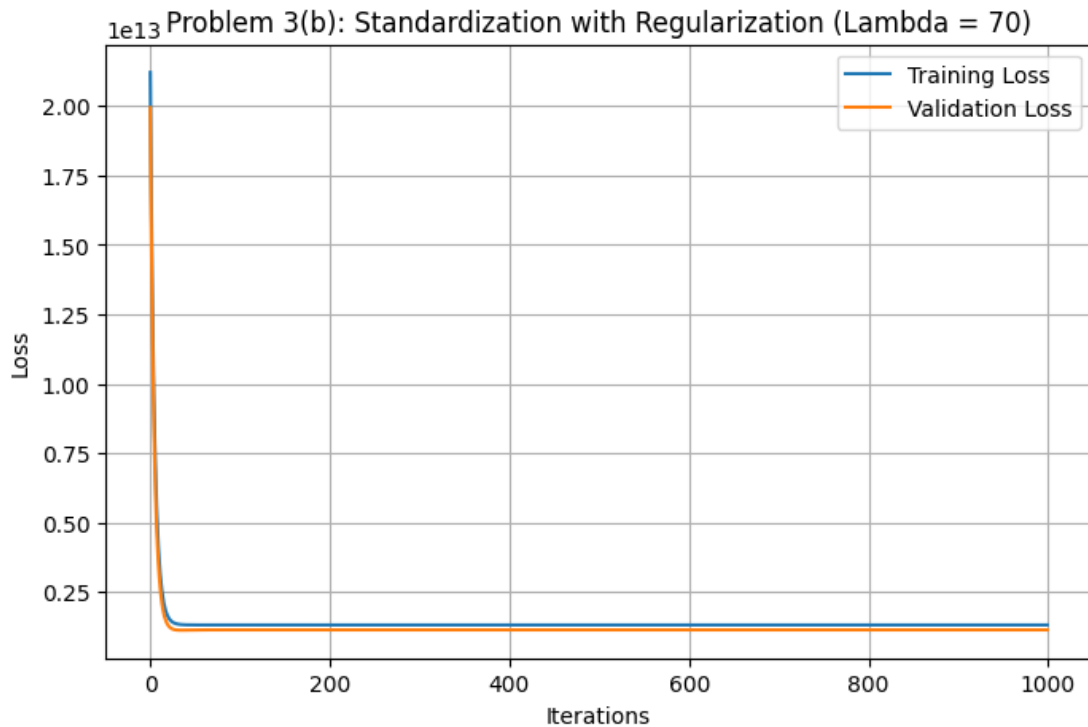


Figure 10. Problem 3(b): standardized 11-feature model with L2 regularization, $\lambda = 70$.

The validation MSE improved from 1.15591×10^{12} in Problem 2(b) to 1.14107×10^{12} in Problem 3(b), a reduction of approximately 1.28%. The area, bathrooms, stories, basement, hot-water heating, air-conditioning, parking, and preferred-area coefficients were smaller in the regularized model. As in Problem 3(a), sufficiently large λ values increased validation error, showing the bias introduced by excessive regularization.

8. Overall Results and Conclusions

Experiment	Scaling / penalty	α	λ	Training loss	Validation MSE
1(a)	None	1e-9	0	3.3801e12	3.2188e12
1(b)	None	1e-9	0	3.3800e12	3.2188e12
2(a)	Min-Max	0.02	0	1.6256e12	1.5094e12
2(a)	Standardized	0.05	0	1.5221e12	1.5965e12
2(b)	Min-Max	0.03	0	1.1806e12	1.1127e12
2(b)	Standardized	0.05	0	1.1485e12	1.1559e12
3(a)	Standardized + L2	0.05	75	1.7121e12*	1.5713e12
3(b)	Standardized + L2	0.05	70	1.3186e12*	1.1411e12

*Problem 3 training losses include the L2 penalty and therefore are not directly comparable with the ordinary training MSE values in Problems 1 and 2.

The experiments show three main results. First, raw housing features caused gradient descent to diverge for the required learning rates of 0.01–0.10 because the input variables had very different numerical scales. A learning rate as small as 10^{-9} stabilized training but made convergence extremely slow. Second, both Min-Max normalization and standardization eliminated this instability. Standardization converged especially quickly and produced the lowest training MSE within the selected 1,000-iteration comparisons, while the selected Min-Max runs had lower validation MSE in Problems 2(a) and 2(b). Third, moderate L2 regularization slightly improved held-out performance for both standardized models. The useful penalty was not arbitrarily large: after the validation minimum, stronger regularization increased error.

Overall, the homework demonstrates that gradient descent depends strongly on feature scale and learning rate, and that regularization introduces a tradeoff between fitting the training data and controlling parameter magnitude. The final 11-feature regularized experiment used $\lambda = 70$ and achieved a validation MSE of approximately 1.1411×10^{12} .